Project Title:

AI Based Cleanliness Monitoring System

1. Problem Domain Identification and Analysis

Domain→ This project belongs to the domain of Computer Vision and Artificial Intelligence applied to Public Safety, Facility Management, and Smart Surveillance. It specifically applies deep learning based image analysis to real time monitoring of floor surfaces in public spaces such as offices, hospitals, malls, railway stations, schools, and industrial premises.

Problem Statement→ places and workplaces require continuous monitoring of floor cleanliness and other things like liquid spills, garbage, and stains. Manual inspection is slow, inconsistent, dependent on staff availability . This delay in detection increases the risk of slips, falls and also increases the operational cost of maintaining cleanliness in large facilities.
There is a need for an automated system that can continuously observe floor areas through cameras, detect the presence of liquid spills, garbage, or stains using artificial intelligence, and immediately generate an alert so that the concerned staff can respond without delay

Why This Problem Is Significant → Floor spills, garbage, and stains are not just a cleanliness issue, they are a real safety risk because slipping on a wet floor can cause serious injury to anyone walking by. Right now most places only clean when someone notices the mess or during a fixed schedule, so the problem sits there for a long time before anyone acts. This delay is dangerous in hospitals, schools, and busy public places where people are moving fast and not always looking down. Hiring people to check floors all day is also costly and hard to manage for big buildings. So an AI system that can catch these hazards early and alert staff right away can prevent accidents and also save cleaning cost and effort.

Project Scope → The project proposes the design and development of an ai based system that uses camera input to continuously analyzes floor and classify them into categories such as clean floor, liquid spill, garbage, and stain. When a problem is detected with best confidence score , the system generates a real time alert to the main staff, through a dashboard notification, sound alert, or messaging service such as an email or chat based alert.

Objectives→ to design a deep learning model that can accurately predict liquid spills, garbage and stains on the surfaces from the camera inages. Also to generate a real time alert within short time window once problem is confidently detected. To reduce the dependencey on the manual floor inspection and preventive facility maintenance.

2. Review of Existing Systems and Project Feasibility

Existing Work	Summary	Limitation
Deep learning based dirt detection for floor cleaning robots (ACIN dataset, YOLOv5 based study)	Uses synthetic and real floor images to detect solid dirt and liquid dirt for robotic floor cleaners. Reported a mean average precision of 0.874 for solid dirt detection.	Focuses mainly on dirt for cleaning robots, not on public safety alerting; struggles with stains being confused as dirt.
Thermal and RGB based spill detection research	Combines thermal and RGB cameras to detect water and liquid spills, since plain RGB cameras often fail to distinguish clear liquid from the floor surface.	Requires additional thermal camera hardware, increasing cost and complexity.
Garbage and litter detection datasets and models (for example TACO, Roboflow garbage detection, YOLO based waste detection studies)	Provide annotated images of litter and garbage for object detection; some recent studies report waste detection accuracy above 90 percent using YOLO based models.	Designed for outdoor waste and litter management, not specifically for indoor floor hazard monitoring or spill and stain detection.
Commercial robotic vacuum and floor cleaning systems	Some robotic cleaners include basic dirt sensing to decide cleaning intensity in a given area.	Built for autonomous cleaning action, not for alerting human staff about hazards for safety purposes.

Identification Gap →Existing systems generally address a single problem tyle isolation , either solid dirt for cleaning robots or litter for outer waste managemet or spill detection , garbage detection, and stain detection into a single monitoring pipeline . This project aims to fill that gap by building a unified detection and alerting system for indoor and semi outdoor areas.

Feasibility →
The projet is technically feasible , Proven deep learning architectures such as YOLO based object detection models and segmentation models such as U-Net and DeepLabV3 are freely available and well documented for similar detection tasks.
Public datasets relevant to this problem already exist and can be used for pretraining, including the ACIN floor dirt dataset, a synthetic floor dirt and spill dataset, and litter detection datasets such as TACO and Roboflow based garbage detection datasets. The project can be completed in the planned milestones using an incremental approach, starting with a single hazard type such as garbage detection, followed by liquid spill detection, and finally combining all classes into one pipeline.

Team members –

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